

1. (10%) Order the following function by growth rate in increasing order:

$$n^2, n \log(\log n), n \log(n^2), n^n$$

$$n^2 = n \cdot n = n \log(n^n)$$

$$\text{當 } n > 2 \text{ 時 } \log n < n^2 < n^n$$

$$\text{故 } n \log(\log n) < n \log(n^2) < n^2 < n^n$$

2. (10%) Ackerman's function $A(m, n)$ is defined as:

$$A(m, n) = \begin{cases} n + 1, & \text{if } m = 0 \\ A(m - 1, 1), & \text{if } n = 0 \\ A(m - 1, A(m, n - 1)), & \text{otherwise} \end{cases}$$

What is the value of $A(3, 4)$?

$$A(0, n) = n + 1$$

$$A(1, n) = A(0, A(1, n - 1)) = A(0, n - 1) + 1, \quad n > 0$$

$$A(1, 0) = A(0, 1) = 2$$

$$\Rightarrow A(1, n) = n + 2$$

$$A(2, n) = A(1, A(2, n - 1)) = A(1, n - 1) + 2, \quad n > 0$$

$$A(2, 0) = A(1, 1) = 3$$

$$\Rightarrow A(2, n) = 2n + 3$$

$$A(3, n) = A(2, A(3, n - 1)) = 2A(3, n - 1) + 3, \quad n > 0$$

$$A(3, 0) = A(2, 1) = 2 \cdot 1 + 3 = 5$$

$$A(3, 1) = 2A(3, 0) + 3 = 2(2 + 3) + 3 = 13 = 2^2 + 3(1 + 2)$$

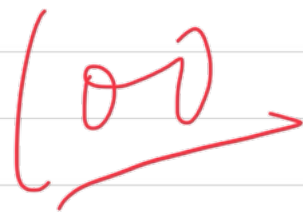
$$A(3, 2) = 2A(3, 1) + 3 = 2[2^2 + 2 \cdot 3 + 3] + 3 = 29 = 2^3 + 3(1 + 2 + 2^2)$$

$$A(3, 3) = 2A(3, 2) + 3 = 2[2^3 + 2^2 \cdot 3 + 2 \cdot 3 + 3] + 3 = 61 = 2^4 + 3(1 + 2 + 2^2 + 2^3)$$

$$A(3, 4) = 2A(3, 3) + 3 = 2^5 + 3 \sum_{k=0}^4 2^k = 125$$

$$\begin{aligned} \Rightarrow A(3, n) &= 2^{n+1} + 3 \sum_{k=0}^n 2^k = 2^{n+1} + 3 \cdot \frac{2^{n+1} - 1}{2 - 1} \\ &= 2^{n+1} [1 + 3] - 3 \\ &= 2^{n+3} - 3 \end{aligned}$$

$$A(3, 4) = 2^7 - 3 = 128 - 3 = 125$$



3. (12%) Show that the following statements are correct:

(a) $5n^2 - 6n = \Theta(n^2)$

(b) $n! = O(n^n)$

(a) 當 $n \geq 2$ 時, 有 $3n^2 \leq 5n^2 - 6n \leq 5n^2$
 $\Rightarrow 5n^2 - 6n = \Theta(n^2)$

(b) 當 $n \geq 1$ 時, 有 $n! = 1 \times 2 \times \dots \times n \leq \underbrace{n \times n \times \dots \times n}_{n \text{ 個}} = n^n$
 $\Rightarrow n! = O(n^n)$

4. (12%) Show that the following statements are incorrect:

(a) $10n^2 + 9 = O(n)$

(b) $3^n = O(2^n)$

(a) 令 $f(n) = 10n^2 + 9$, $g(n) = n$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{10n^2 + 9}{n} = 10n \Rightarrow \text{發散}$$

故 $10n^2 + 9 \neq O(n)$

(b) 令 $f(n) = 3^n$, $g(n) = 2^n$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{3^n}{2^n} = \left(\frac{3}{2}\right)^n \Rightarrow \text{發散}$$

故 $3^n \neq O(2^n)$

5. (16%) (from NTHU 106) Which ones of the following conjectures are correct?

Please justify your answers:

(a) $3n^2 - 100n + 6 = O(n)$

(b) $\log(n!) = \Theta(n \log n)$

(c) $n! = O(n^n)$

(d) $(n+a)^b = \Theta(n^b)$ where a and b are real constants

(e) $(n+a)^b = \Omega(n^{b+1})$ where a and b are real constants

(a) 當 $n \geq 50$ 時有 $3n^2 - 100n + 6 \geq n^2$
 $\Rightarrow 3n^2 - 100n + 6 = \Omega(n^2)$
 n^2 增長的速率較 n 快
 $\Rightarrow 3n^2 - 100n + 6 \neq O(n)$

(b) $\log(n!) = \sum_{k=1}^n \log(k) \leq \sum_{k=1}^n \log(n) = n \log(n)$

當 $n \geq 1$ 時有 $\frac{1}{2} n \log n \leq \log(n!) \leq n \log n \Rightarrow \log(n!) = \Theta(n \log n)$

(c) $n! = \prod_{k=1}^n k \leq \prod_{k=1}^n n = n^n \Rightarrow n! = O(n^n)$

(d) $\lim_{n \rightarrow \infty} (n+a)^b = \lim_{n \rightarrow \infty} \sum_{k=0}^b C_k^b n^k a^{b-k}$
 $\approx n^b$
 $\Rightarrow (n+a)^b = \Theta(n^b)$

(e) 由 (d) 知 $(n+a)^b = \Theta(n^b) \Rightarrow (n+a)^b = O(n^b)$
 n^b 增長的速率較 n^{b+1} 慢
故 $(n+a)^b \neq \Omega(n^{b+1})$

Ans: (b), (c), (d) 正確

6. (20%) Fill the step count table (yellow regions) for the following program:

$\text{let } n = \text{MAX_SIZE}$

Statement	s/e	Frequency	Total Steps
<code>void mult(int a[][MAX_SIZE], int b[][MAX_SIZE], int c[][MAX_SIZE])</code>	0	0	0
<code>{</code>	0	0	0
<code>int i, j, k;</code>	0	0	0
<code>for (i = 0; i < MAX_SIZE; i++)</code>	1	$n+1$	$n+1$
<code>for (j = 0; j < MAX_SIZE; j++) {</code>	1	$n(n+1)$	n^2+n
<code>c[i][j] = 0;</code>	1	n^2	n^2
<code>for (k = 0; k < MAX_SIZE; k++)</code>	1	$n^2(n+1)$	n^3+n^2
<code>c[i][j] += a[i][k] * b[k][j];</code>	1	n^3	n^3
<code>}</code>	0	0	0
<code>}</code>	0	0	0
Total			

$\rightarrow 2n^3 + 3n^2 + 2n + 1$

7. (10%) Given an four-dimensional array $A[100][200][300][400]$, if the address of $A[3][20][32][45]$ is $\alpha + X$, what is the value of X ?

Note: (1) α is the address of $A[0][0][0][0]$

(2) Only consider row major here.

$$\begin{aligned}
 \alpha + X &= \alpha + 3 \cdot 200 \cdot 300 \cdot 400 + 20 \cdot 300 \cdot 400 + 32 \cdot 400 + 45 \\
 &= \alpha + 72000000 + 2400000 + 12800 + 45 \\
 &= \alpha + 74,412,845
 \end{aligned}$$

8. (10%) (from NTCU 107) Consider the following function. Please derive the time complexity of function foo using recurrence relation. Please show your derivation step by step.

`int i[n];` /* Assume that the values of all integers in i[n] have been initialized */

	s/e	頻率	Total Steps
<code>int foo(int a, int b, int c) {</code>			
<code>int d;</code>			
<code>if (a > b) return -1;</code>			
<code>d = (a + b) / 2;</code>	1	$\log n$	$\log n$
<code>if (c == i[d])</code>	1	$\log n$	$\log n$
<code>return d;</code>	1	1	1
<code>else if (c > i[d])</code>	1	$\log n$	$\log n$
<code>return foo(d + 1, b, c);</code>	1	$\log n$	$\log n$
<code>else</code>	1	$\log n$	$\log n$
<code>return foo(a, d - 1, c);</code>	1	$\log n$	$\log n$
<code>}</code>			

$6\log n + 1$

$\Rightarrow$ 時間複雜度為 $O(\log n)$

